Table D1: Forecast accuracy with five and ten variance lags

	MSE		MAE	
Model	5 lags	10 lags	5 lags	10 lags
Equity-only VAR	0.368531	0.365940	0.464175	0.463306
Equity-only LVAR	0.334215	0.328510	0.442837	0.440629
LVAR: oil and gold	0.336611	0.330827	0.443919	0.442912
LVAR: natural gas	0.332670	0.328684	0.442343	0.441537
LVAR: oil, natural gas and gold	0.334549	0.330157	0.443412	0.442896
LVAR: all four commodities	0.334687	0.330471	0.443561	0.442872

Notes: all specifications forecast the same 783 ES log-RV5 outcomes from 13 July 2023 to 31 August 2026, using an expanding estimation sample. Lower MSE and MAE indicate smaller errors. Every LVAR includes the same three leverage inputs (downside controls) based on 1/5/22-observation equity returns; these windows remain fixed when the variance lag order changes. The equity-only VAR and LVAR are single-variable autoregressions, labelled consistently with the main report. "All four" means oil, corn, gold and natural gas. Source: own calculations in `VAR_QTFE.ipynb`, Section 10, final-data run of 13 October 2026.

The leverage inputs reduce equity-only MSE by 9.31% at five lags and 10.23% at ten lags. At ten lags, every reported commodity extension has higher MSE than the equity-only LVAR. The small natural-gas advantage at five lags therefore does not persist in this check.